

# Bach Phan-Tat

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Computational linguist and data scientist developing interpretable methods for detecting and analysing CHANGE. My current research focuses on lexical and conceptual change using NLP techniques and (diachronic) corpus data. I am also extending this work towards financial time-series analysis and potentially other domains that CHANGE.

## WORK EXPERIENCE

**The Digital Humanities Institute, University of Sheffield**      **Sep 2026 – Nov 2026**  
*Visiting Research Scholar*      *Sheffield, England*

**Vbee (speech technology and conversational AI company)**      **Feb 2023 – Aug 2024**  
*Junior Research Engineer*      *Hanoi, Vietnam*  
*Project: HUST Chatbot (a conversational AI for Hanoi University of Science and Technology)*

- Developed and evaluated machine-learning and BERT-based models for intent/domain classification, toxicity detection, and name-based classification, achieving over 85% accuracy on held-out evaluation sets.
- Designed data-sampling and human-evaluation procedures, applying statistical methods to assess dataset quality, annotation reliability, and model performance.
- Built an LLM-assisted chatbot evaluation pipeline, including evaluation criteria, prompt design, automated response scoring, and systematic analysis of model outputs.
- Developed language-data pipelines covering web crawling, information extraction, LLM-assisted data generation, preprocessing, annotation, and quality control for text and speech datasets.
- Contributed to the development of linguistic resources, and rule-based components for speech and language tasks, including speech-to-text, text-to-speech, and part-of-speech parsing.

**what3words (location-mapping technology company)**      **Jul 2023 – Aug 2023**  
*Remote Research Consultant*      *London, England*  
*Project: Vietnamese Automatic Speech Recognition system (in collaboration with VinFast)*

- Conducted systematic linguistic error analysis of Vietnamese ASR outputs, categorising recurring errors involving phonetic confusions, lexical ambiguity, and contextual misrecognition.
- Translated error-analysis findings into data- and model-level recommendations.

## ACADEMIC & PROFESSIONAL SERVICE

### Supervising and Mentoring

- Advised a Master's student at Lancaster University on a Linguistics dissertation, which is awarded a Distinction (76%). 2024.
- Mentored applicants in postgraduate and fellowship applications, with subsequent admissions to the Linguistics PhD programme at Lancaster University (2024), the YSEALI Fellowship (2024), the Stipendium Hungaricum Scholarship (2026).

### Reviewing

- Reviewer for Cognitive Linguistic Studies. 2026.
- Reviewer for Linguists' Day, Linguistic Society of Belgium (LSB). 2025.

## Organising

- Co-organised the PhD Day for the Department of Linguistics at KU Leuven. 2026.

## PUBLICATIONS

### Book chapters (peer-reviewed)

- Sofía Aguilar-Valdez<sup>\*</sup>, **Bach Phan-Tat**<sup>\*</sup>, Dirk Speelman, Dirk Geeraerts, and Stefania Degaetano-Ortlieb. 2026. Discursive parallels of the chemical revolution: topic modelling and distributional analysis. In *Understanding Science with Large Language Models? Potentials for the History, Philosophy, and Sociology of Science* (open access, transcript). (Book chapter / edited volume). <sup>\*</sup>Equal contribution.

### Conference and workshop proceedings (peer-reviewed)

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. ReFRAME or Remain: Unsupervised Lexical Semantic Change Detection with Frame Semantics. *Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (\*SEM 2026)*, *ACL 2026*.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Transparent Semantic Change Detection with Dependency-Based Profiles. *Proceedings of the 6th International Workshop on Computational Approaches to Historical Language Change (LChange'26)*, *EACL 2026*.
- Khai Le-Duc, Tuyen Tran, **Bach Phan-Tat**, et al. 2025. MultiMed-ST: Large-scale Many-to-many Multilingual Medical Speech Translation. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*.
- Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, **Bach Phan-Tat**, Minh-Huong Ngo, Thanh Nguyen-Tang, Truong-Son Hy. 2025. MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Top 15% of accepted papers.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, Le Duy, Long Vo-Dang, Truong-Son Hy. 2025. Sentiment Reasoning for Healthcare. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Oral presentation.

### Journal articles (peer-reviewed)

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. Evaluating the Evaluator: Problems with SemEval-2020 Task 1 for Lexical Semantic Change Detection. *Journal of Open Humanities Data*.
- Catherine Wong, **Bach Phan-Tat**, Susan Fitzmaurice. 2026. Methods, Data, and Conceptual Change: Reflections from Two Quantitative Diachronic Case Studies. *Journal of Open Humanities Data*.

### Pre-prints and under-review articles

- Duc Cao-Dinh, Khai Le-Duc, Anh Dao, **Bach Phan-Tat**, Chris Ngo, Duy M. H. Nguyen, Nguyen X. Khanh, Thanh Nguyen-Tang. 2025. Audio-3DVG: Unified Audio - Point Cloud Fusion for 3D Visual Grounding. (Preprint).

## PRESENTATIONS

### Conference presentations

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Underlying Dimensions of Semantic Change. Digital Humanities Congress 2026.
- **Bach Phan-Tat**, Kris Heylen, and Stefano De Pascale. 2026. Lexical Semantic Change Detection? A Good Parser Might Be Enough! 1st LOT Conference.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. SynFlow: Continuous Semantic Change Analysis via Dependency Co-occurrences. ICAME 47.
- **Bach Phan-Tat**, Dirk Geeraerts, and Dirk Speelman. 2026. Conceptual Change during the Chemical Revolution: Air, Acid, and Water in the Royal Society Corpus. ICAME 47.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. From Parsers to Prompts: Combining AI and Linguistics for Interpretable Language Evolution. AI in Language Evolution Workshop, Evolang 2026.
- **Bach Phan-Tat**. 2026. Slot-Filler Distributional Changes in Times of Scientific Debate: A Case Study of Air. Quantitative Diachronic Linguistics and Cultural Analytics.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. 2024. Sentiment Reasoning for Healthcare. Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond Workshop, NeurIPS 2024.

#### Invited talks & seminars

- **Bach Phan-Tat**. 2026. Underlying Dimensions of Semantic Change. NLP & text mining seminar organised by Leuven.AI.
- **Bach Phan-Tat**. 2026. SynFlow: Continuous Semantics Change Analysis via Dependency Co-occurrences. Data in Historical Linguistics seminar series.

## EDUCATION

### KU Leuven

*PhD, Computational Linguistics*

**Sep 2024 – Present**

*Leuven, Belgium*

Thesis: **Underlying Dimensions in Conceptual Change**

#### Selected Achievements

- Developed a general, lightweight, and interpretable framework for lexical semantic change detection and analysis, using off-the-shelf parsers or LLMs to extract structured linguistic patterns and Jensen-Shannon Divergence to quantify diachronic distributional shifts.
- Developed SynFlow, a Python package implementing the framework through syntactic co-occurrence extraction, temporal comparison, and incremental clustering of diachronic type embeddings.
- Implemented syntactic pattern extraction and structuring using breadth-first and depth-first traversal, n-ary tree search, and linguistically motivated rule-based filtering.
- Built automated pipelines for LLM-assisted linguistic annotation and for cleaning, lemmatising, and parsing large corpora.
- Optimised computational workflows through parallel and batch processing, reducing processing time by 80%.
- Maintained and extended NephoSem, a lexical-semantics toolkit developed by the QLVL research unit, adding support for multiple embedding representations in downstream analyses.

Thesis supervisors: Prof. Dr. Dirk Speelman, Prof. Dr. Dirk Geeraerts, Dr. Kris Heylen

### WorldQuant University

*MSc, Financial Engineering*

**Oct 2025 – Present**

*Remote*

Finished modules: Financial Market, Financial Data, Financial Econometrics

### University of Stirling

**Sep 2021 – Sep 2022**

Thesis: **The structure of *thời gian*: A semantic analysis of the Vietnamese lexeme *thời gian* based on Principled Polysemy**

### Selected Achievements

- Randomly sampled, crawled, cleaned, and compiled a specialised corpus of 500 Vietnamese news articles using Python and #LancsBox.
- Identified seven senses of the Vietnamese lexeme *thời gian* ('time') and traced their semantic development using the Principled Polysemy framework.

Thesis supervisor: Dr. Andrew Smith

Thesis grade: 80% (Distinction; class rank: 1st)

Average grade: 78% (Distinction; class rank: 1st)

### Academy of Finance

Aug 2016 – Aug 2020

BA, Corporate Finance

Hanoi, Vietnam

Relevant modules: Linear Algebra, Calculus, Probability and Statistics, Econometrics

## TECHNICAL SKILLS

### Programming & Computing

- Programming languages (ordered by proficiency): Python (NumPy, Pandas, Matplotlib, scikit-learn, spaCy, PyTorch, Hugging Face), R, Bash, LaTeX
- Parallel, batch, and GPU-enabled computing workflows
- Databases: MongoDB, SQL
- Git/GitHub

### Data Science

- Exploratory data analysis, data cleaning and (interactive) data visualisation
- Machine learning and deep learning for NLP, time-series, structured data and computer vision
- Quantitative research design, experimental design, and A/B testing
- LLM-based annotation and evaluation workflows
- Mathematics for data science: linear algebra, multivariate calculus, statistics

### Linguistics & Corpus Methods

- Corpus tools: CQPweb, #LancsBox, Sketch Engine, AntConc
- Linguistic analysis: phonetics, morphology, syntax, semantics
- Linguistic annotation, transcription, and corpus preparation

## AWARDS AND HONOURS

- **2026: First Place, Databricks Challenge** (US\$1,000), Hack-Nation's 4th Global AI Hackathon; 3,000+ participants from 300+ universities across 65+ countries.
- **2024: Marie Skłodowska-Curie Actions Doctoral Fellowship** (fully funded for three years), Horizon Europe; awarded to 10 out of 284 applicants worldwide.
- **2022: Research-Based Learning Prize for the best Master's dissertation in Literature and Languages** (GBP 225), University of Stirling; the first recipient in the history of the Linguistics department.
- **2021: Postgraduate Scholarship** (GBP 4,000), University of Stirling.

## **SELECTED ADDITIONAL TRAINING**

### **Data Science**

- Inferential Statistics (The University of Amsterdam)
- Applied Data Science Lab (WorldQuant University)
- Deep Learning (Neuromatch Academy)
- Applied AI Lab: Deep Learning for Computer Vision (WorldQuant University)

### **Linguistics**

- Corpus Linguistics: Method, Analysis, Interpretation (Lancaster University)
- Corpus Linguistics Summer School (Birmingham University)

### **Financial Engineering**

- Foundations of Financial Engineering (WorldQuant University)

### **Research Methodology**

- Quantitative Method (The University of Amsterdam)
- Qualitative Research Methods (The University of Amsterdam)

## **LANGUAGES**

- Vietnamese: Native
- English: IELTS 8.5 – C2
- Dutch: A2